

HANDWRITTEN DIGIT RECOGNIZER

Project Report

Abstract

This project implements a Handwritten Digit Recognizer using Deep Learning and the MNIST dataset. A Convolutional Neural Network (CNN) was trained to classify handwritten digits (0–9). The model was integrated into a Streamlit web application that allows users to upload digit images and receive predictions with confidence scores.

Objectives

- Develop an accurate handwritten digit classification model.
- Train and evaluate a CNN using the MNIST dataset.
- Build an interactive Streamlit-based user interface.
- Demonstrate real-time digit prediction from uploaded images.

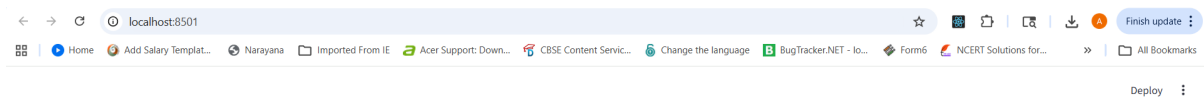
Technology Stack

Python, TensorFlow/Keras, NumPy, Pandas, OpenCV, Matplotlib, Streamlit

System Architecture

The workflow consists of dataset preprocessing, CNN model training, evaluation, model saving, and deployment through a Streamlit application.

Application Home Page



Handwritten Digit Recognizer

Upload Digit Image

Upload

200MB per file • PNG, JPG

Model Training

The CNN model was trained for 5 epochs. Training and validation accuracy improved steadily, demonstrating successful learning and generalization.

```
PS C:\Users\Aaryani\Desktop\Handwritten-Digit-Recognizer> python train_model.py
ugin.
Epoch 1/5
1688/1688 ██████████ 26s 15ms/step - accuracy: 0.9522 - loss: 0.1506 - val_accuracy: 0.9752
- val_loss: 0.0828
Epoch 2/5
1688/1688 ██████████ 47s 18ms/step - accuracy: 0.9852 - loss: 0.0476 - val_accuracy: 0.9887
- val_loss: 0.0383
Epoch 3/5
1688/1688 ██████████ 24s 14ms/step - accuracy: 0.9888 - loss: 0.0356 - val_accuracy: 0.9867
- val_loss: 0.0522
Epoch 4/5
1688/1688 ██████████ 17s 10ms/step - accuracy: 0.9915 - loss: 0.0261 - val_accuracy: 0.9915
- val_loss: 0.0299
Epoch 5/5
1688/1688 ██████████ 18s 11ms/step - accuracy: 0.9932 - loss: 0.0212 - val_accuracy: 0.9900
- val_loss: 0.0365
313/313 ██████████ 1s 4ms/step - accuracy: 0.9915 - loss: 0.0310
Accuracy: 0.9915000200271606
```

Training Highlights:

- Final Training Accuracy: 99.32%
- Validation Accuracy: Approximately 99.00%
- Low validation loss indicating strong model performance.

Model Evaluation

```
PS C:\Users\Aaryani\Desktop\Handwritten-Digit-Recognizer> python train_model.py
313/313  1s 3ms/step - accuracy: 0.9917 - loss: 0.0294
Accuracy: 0.9916999936103821
```

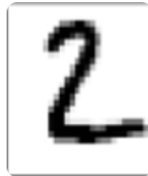
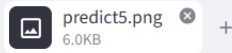
The trained model achieved an overall test accuracy of approximately 99.17%, indicating excellent classification performance on unseen handwritten digit samples.

Application Testing

Uploaded Digit Example - Digit 2

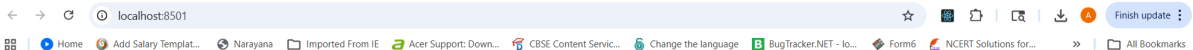
Handwritten Digit Recognizer

Upload Digit Image



Uploaded Image

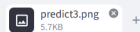
Prediction Example - Digit 9



Deploy

Handwritten Digit Recognizer

Upload Digit Image



Uploaded Image

Predicted Digit: 9 (100.00% confidence)

Results and Analysis

The developed system successfully recognizes handwritten digits with high accuracy. The Streamlit application provides an intuitive interface for users to upload images and obtain instant predictions. The CNN model demonstrates excellent performance on the MNIST dataset and achieves over 99% test accuracy.

Conclusion

The Handwritten Digit Recognizer project successfully combines deep learning and web deployment technologies. The model achieves high prediction accuracy while maintaining an easy-to-use interface, making it suitable for educational and practical image classification applications.

Future Enhancements

- Deploy on cloud platforms.
- Support real-time canvas drawing.
- Extend recognition to alphabets and symbols.
- Improve robustness for custom handwritten inputs.